

--	--	--	--	--	--	--	--

B.Tech. Degree VII Semester Supplementary Examination
April 2019

ME 15-1702 VIBRATION AND NOISE CONTROL
(2015 Scheme)

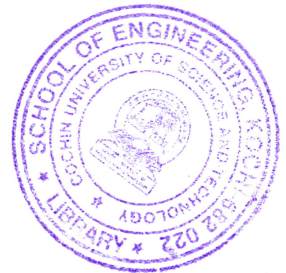
Time: 3 Hours

Maximum Marks: 60

PART A
(Answer ALL questions)

(10 × 2 = 20)

- I. (a) Show that a rotating unbalance system acts as a forced vibration system.
- (b) Write a short note on coulomb damping and its application.
- (c) What do you understand by a support excited motion?
- (d) Explain the working principle of accelerometer.
- (e) What do you mean by a torsionally equivalent shaft?
- (f) What are mode shapes?
- (g) What do you mean by whirling of shaft?
- (h) What is a continuous system?
- (i) What are the physiological effects of noise?
- (j) Write a short note on noise exposure limits.



PART B

(4 × 10 = 40)

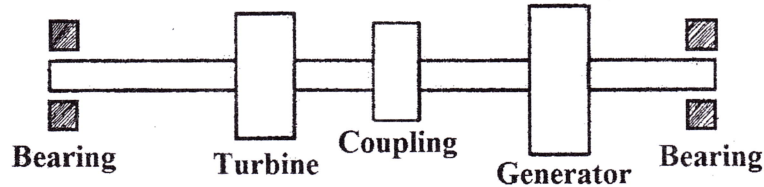
- II. (a) Explain in detail about under damped, critically damped and over damped systems. (4)
- (b) A door 200 cm high, 75 cm wide, 4 cm thick and having a mass of 35 kg is fitted with an automatic door closer. The door opens against a spring with a torsional stiffness of 10 N.cm/radian. If the door is opened 90° and released, how long it will take the door to be within 1° of closing? Assume the return spring of the door to be critically damped. (6)

OR

- III. (a) A machine of mass 75 kg is mounted on springs and is fitted with a dashpot to damp out vibrations. There are three springs each of stiffness 10 N/mm and it is found that the amplitude of vibration diminishes from 38.4 mm to 6.4 mm in two complete oscillations. Assuming that the damping force varies as the velocity, determine: (i) resistance of the dashpot at unit velocity (ii) the ratio of the frequency of the damped vibration to the frequency of the un-damped vibration (iii) the periodic time of the damped vibration. (5)
- (b) A single cylinder vertical petrol engine of total mass 300 kg is mounted upon a steel chassis frame and causes a vertical static deflection of 2 mm. The reciprocating parts of the engine have a mass of 20 kg and move through a vertical stroke of 150 mm with simple harmonic motion. A dashpot is provided whose damping resistance is directly proportional to the velocity and amounts to 1.5 kN per meter per second. Considering that the steady state of vibration is reached; determine: (i) the amplitude of forced vibrations, when the driving shaft of the engine rotates at 480 rpm (ii) the speed of the driving shaft at which resonance will occur. (5)

(P.T.O.)

- IV. Obtain torsional natural frequencies of a turbine-coupling-generator rotor system as shown in figure. The rotor is assumed to be supported on frictionless bearings. The polar mass moment of inertia of the turbine, coupling and generator are $I_{p1} = 25 \text{ kg-m}^2$, $I_{p2} = 5 \text{ kg-m}^2$ and $I_{p3} = 50 \text{ kg-m}^2$, respectively; and these are assumed to be thin discs. Take the modulus of rigidity of the shaft as $G = 0.8 \times 10^{11} \text{ N/m}^2$. Assume the shaft diameter to be uniform throughout and is equal to 0.2 m and the length of shafts between the bearing-turbine-coupling-generator-bearing are 1 m each so that the total span is 4 m. Consider the shaft to have negligible mass. (10)



OR

- V. (a) Explain the working principle of seismometer. Also explain the response curve with operating limits. (4)
- (b) Using Lagrange's equation, determine the equations of motion of a two degrees of freedom vibrating system with a suitable example. (6)
- VI. A vertical steel shaft 15 mm diameter is held in long bearings 1 metre apart and carries at its middle a disc of mass 15 kg. The eccentricity of the centre of gravity of the disc from the centre of the rotor is 0.30 mm. The modulus of elasticity for the shaft material is 200 GN/m^2 and the permissible stress is 70 MN/m^2 . Determine (i) the critical speed of the shaft (ii) the range of speed over which it is unsafe to run the shaft. Neglect the mass of the shaft. (10)
- OR
- VII. Derive the equation for axial vibration of a rod with a solution procedure. (10)
- VIII. (a) Explain the following terms (i) sound pressure level (ii) sound intensity level (iii) sound power level (iv) loudness (v) octave bands. (5)
- (b) Six machines operating individually, makes sound pressure levels of 78 dB, 81 dB, 81 dB, 79 dB, 72 dB and 65 dB respectively. Determine the overall sound pressure level when all of them operate simultaneously. (5)
- OR
- IX. (a) What are the various effects of continuous exposure of noise on human body? (5)
- (b) Explain in detail about noise control. (5)
